Clicker key: three class sets

Each deck carries three question sets. Pick the set on the deck's title bar (Class set A, B, C) and the deck swaps its clicker questions and room code; the dashboard has the same rooms across the top. Use a different set for each class so answers do not travel between sections. Concept clickers: 30 s, confidence points, brain dump score. ID clickers: 15 s, one tap, lab score.

Respiratory system · rooms RESP4 (set A), RESP4B, RESP4C

SET A · ROOM RESP4 · KEY: ADBCADBCBCADCBADACBDC			ANSWER	
1	CONCEPT	A patient cannot seal off the nasopharynx while swallowing, so liquid comes out the nose. Which structure is not doing its job?	A	Soft palate and uvula
2	ID	Name the structure at the pin.	D	Vocal folds (true vocal cords)
3	CONCEPT	A child's adenoids are enlarged and block the back of the nose. Which tonsil is enlarged, and in which region of the pharynx?	B	Pharyngeal tonsil, nasopharynx
4	ID	Name the structure at the pin.	C	Larynx
5	CONCEPT	Feeling down the front of your neck from the chin, you hit a hard ridge, then a soft gap, then a firm ring. In order, what did you feel?	A	Thyroid cartilage, cricothyroid ligament, cricoid cartilage
6	ID	Name the structure at the pin.	D	Trachea
7	CONCEPT	Which paired cartilages pivot to open and close the glottis, and where do they sit?	B	Arytenoid cartilages, on the back of the cricoid cartilage
8	CONCEPT	During suctioning, the catheter tip touches a ridge at the sternal angle and the patient coughs violently. What did it touch?	C	Carina
9	ID	Name the structure at the pin.	B	Right primary bronchus
10	CONCEPT	Air has just left a lobar bronchus. Which airway does it enter next, and how many of them does the right lung have?	C	Segmental bronchus, ten
11	CONCEPT	An x-ray shows a mass in the lung lobe that sits above the oblique fissure and has no horizontal fissure. Which lobe and which lung?	A	Superior lobe, left lung
12	ID	Name the structure at the pin.	D	Apex
13	CONCEPT	In the hilum of the left lung, which structure is highest, and which are lowest?	C	Highest: pulmonary artery. Lowest: pulmonary veins
14	ID	Name the structure at the pin.	B	Pulmonary arteries
15	CONCEPT	Gas leaving an alveolus crosses three layers to reach the blood. Which list has them in order from air to blood?	A	Type I cell, fused basement membranes, capillary endothelium
16	ID	Name the structure at the pin.	D	Pulmonary venules
17	CONCEPT	On the trachea slide, the tall cells with hairlike projections and the scattered pale cells between them form which epithelium, and what do they do?	A	Pseudostratified ciliated columnar; mucus traps debris and cilia sweep it toward the pharynx
18	ID	Name the structure at the pin.	C	Esophagus
19	CONCEPT	Which structure passes through the diaphragm at the lowest level, and what travels with it?	B	Aorta, with the thoracic duct
20	ID	Name the structure at the pin.	D	Inferior vena cava
21	CONCEPT	A patient has a disorder of the conducting zone: the airways narrow because their smooth muscle contracts and there is no cartilage to hold them open. Which disorder and which structure?	C	Asthma, bronchioles

SET B · ROOM RESP4B · KEY: DBACBDCADCBADACBACDBA			ANSWER	
1	CONCEPT	A patient is asked to say 'ah' and the clinician watches the uvula rise. Which structure is the uvula part of, and which region of the pharynx does it help close?	D	The soft palate; it closes the nasopharynx
2	CONCEPT	A child has a sore throat and the clinician sees two swollen masses at the sides of the throat, between the arches behind the tongue. Which tonsils are these?	B	Palatine tonsils
3	ID	Name the structure at the pin.	A	Uvula
4	CONCEPT	Which laryngeal cartilage is the only complete ring, and what sits directly on top of its wide back plate?	C	Cricoid cartilage; the arytenoid cartilages
5	ID	Name the structure at the pin.	B	Cricoid cartilage
6	CONCEPT	Which cartilage of the larynx is made of elastic cartilage rather than hyaline, and why does that fit its job?	D	Epiglottis; it must bend over the inlet and spring back
7	ID	Name the structure at the pin.	C	Vocal folds
8	CONCEPT	Why does an inhaled object usually end up in the right lung rather than the left?	A	The right main bronchus is wider, shorter and more vertical
9	CONCEPT	As the bronchial tree branches, what happens to cartilage and to smooth muscle?	D	Cartilage decreases to none, smooth muscle increases
10	ID	Name the structure at the pin.	C	Laryngeal prominence
11	CONCEPT	A patient's left lung is examined. How many lobes and fissures should it have, and what does it have that the right lung does not?	B	Two lobes, one fissure, a cardiac notch and lingula
12	ID	Name the structure at the pin.	A	Superior lobe
13	CONCEPT	At the hilum of the lung, which vessel carries oxygen-poor blood in, and which carry oxygen-rich blood out?	D	Pulmonary artery in; pulmonary veins out
14	ID	Name the structure at the pin.	A	Visceral layer
15	CONCEPT	Which alveolar cell makes up most of the wall, and what is its shape and job?	C	Type I cell; thin and flat; gas crosses it
16	ID	Name the structure at the pin.	B	Alveolus
17	CONCEPT	Why does the trachea need cilia and goblet cells, while the alveoli have neither?	A	The trachea must clear inhaled debris upward; alveoli must stay thin for gas exchange
18	ID	Name the structure at the pin.	C	Tracheal cartilage (hyaline cartilage)
19	CONCEPT	A spinal injury at C2 stops breathing; an injury at T6 does not. Which nerve and which spinal levels explain this?	D	Phrenic nerve, C3 to C5, to the diaphragm
20	ID	Name the structure at the pin.	B	Aorta
21	CONCEPT	A patient's alveolar walls have broken down and merged into large air spaces, and the damage is permanent. Which disorder?	A	Emphysema

SET C · ROOM RESP4C · KEY: CADBCABDBACDADBCBCDAB			ANSWER	
1	CONCEPT	Food leaves the oral cavity and has to cross one region of the pharynx that air also uses before it reaches the esophagus. Which region carries both, and what lines it?	C	Oropharynx, stratified squamous
2	ID	Name the structure at the pin.	A	Tongue
3	CONCEPT	Which region of the pharynx opens into the auditory tubes, and why does an enlarged tonsil there lead to ear infections?	D	Nasopharynx; the pharyngeal tonsil blocks the tube opening
4	ID	Name the structure at the pin.	B	Vestibular folds (false vocal folds)
5	CONCEPT	A patient has a goiter. Which structure sits just in front of the upper trachea, below the cricoid, and could press on the airway?	C	Thyroid gland
6	ID	Name the structure at the pin.	A	Thyroid cartilage
7	CONCEPT	The open side of each tracheal C ring faces backward. What fills that gap, and what does it allow?	B	Trachealis muscle; the esophagus can bulge forward during a swallow
8	CONCEPT	A nurse measuring tube placement wants the tip to sit above the point where the trachea splits. At what vertebral level, and what surface landmark, is that split?	D	T4 to T5, the sternal angle
9	ID	Name the structure at the pin.	B	Phrenic nerve
10	CONCEPT	Which airway is the last part of the conducting zone, and what makes the next airway part of the respiratory zone?	A	Terminal bronchiole; the respiratory bronchiole has alveoli in its wall
11	ID	Name the structure at the pin.	C	Carina
12	CONCEPT	The narrow top of the lung reaches above a bone in the neck region. Which part of the lung, and above which bone?	D	Apex, above the clavicle
13	ID	Name the structure at the pin.	A	Base
14	CONCEPT	Which layer of pleura can feel pain, and which cannot, and why does that matter for pleurisy?	D	Parietal feels pain (intercostal and phrenic nerves); visceral does not
15	CONCEPT	A pathologist sees dust-laden cells inside the alveoli of a smoker's lung. Which cells, and what do they do?	B	Alveolar macrophages; they engulf debris and microbes
16	ID	Name the structure at the pin.	C	Branches of pulmonary artery
17	ID	Name the structure at the pin.	B	Visceral pleura
18	CONCEPT	On the trachea slide the nuclei sit at different heights but every cell touches the basement membrane. What is this epithelium called, and where else in the airway is it found?	C	Pseudostratified ciliated columnar; nasal cavity and bronchi
19	CONCEPT	Which structure passes through the central tendon of the diaphragm, and at what level?	D	Inferior vena cava, T8
20	ID	Name the structure at the pin.	A	Central tendon
21	CONCEPT	A premature infant's alveoli collapse after every breath. Which cell has not matured, and which disorder is this?	B	Type II cells; infant respiratory distress syndrome

Lower limb · rooms LIMB4 (set A), LIMB4B, LIMB4C

SET A · ROOM LIMB4 · KEY: ADCBADB			ANSWER	
1	CONCEPT	In the femoral triangle, from lateral to medial, what is the order of the nerve, artery and vein?	**A**	Nerve, artery, vein
2	ID	Name the structure at the pin.	**D**	Rectus femoris
3	CONCEPT	The sciatic nerve leaves the pelvis and enters the gluteal region. Through which opening, and in what relation to which muscle?	**C**	Greater sciatic foramen, below piriformis
4	ID	Name the structure at the pin.	**B**	Sciatic nerve
5	CONCEPT	A patient cannot dorsiflex the foot after an injury at the side of the knee, and the foot slaps when walking. Which nerve, and where was it hurt?	**A**	Common fibular nerve, at the neck of the fibula
6	ID	Name the structure at the pin.	**D**	Deep fibular nerve
7	CONCEPT	Where do you feel the posterior tibial pulse, and which nerve runs beside the artery there?	**B**	Behind the medial malleolus, tibial nerve

SET B · ROOM LIMB4B · KEY: BACDBDC			ANSWER	
1	CONCEPT	Which four muscles form the quadriceps, and which one of them also flexes the hip?	**B**	Rectus femoris and three vasti; rectus femoris flexes the hip
2	ID	Name the structure at the pin.	**A**	Femoral artery
3	CONCEPT	Which three muscles are the hamstrings, where do they all originate, and what do they do?	**C**	Biceps femoris, semitendinosus, semimembranosus; ischial tuberosity; flex the knee and extend the hip
4	CONCEPT	Which muscle dorsiflexes and inverts the foot, and which nerve supplies it?	**D**	Tibialis anterior; deep fibular nerve
5	ID	Name the structure at the pin.	**B**	Superficial fibular nerve
6	CONCEPT	Gastrocnemius and soleus share one tendon. What is it called, where does it insert, and what movement do they produce?	**D**	Calcaneal tendon; calcaneus; plantar flexion
7	ID	Name the structure at the pin.	**C**	Calcaneal tendon

SET C · ROOM LIMB4C · KEY: CDABADC			ANSWER	
1	CONCEPT	The femoral triangle has three borders. Which is which?	**C**	Inguinal ligament above, sartorius lateral, adductor longus medial
2	ID	Name the structure at the pin.	**D**	Vastus intermedius
3	CONCEPT	A patient cannot keep the pelvis level when standing on one leg; the opposite hip drops. Which muscles are weak, and which nerve supplies them?	**A**	Gluteus medius and minimus; superior gluteal nerve
4	ID	Name the structure at the pin.	**B**	Semitendinosus
5	CONCEPT	The common fibular nerve splits into two branches. Which compartment does each run, and what does each do?	**A**	Deep: anterior leg, dorsiflexion. Superficial: lateral leg, eversion
6	ID	Name the structure at the pin.	**D**	Fibularis brevis
7	CONCEPT	A deep vein clot forms in the leg. Which veins are deep, and what name do they share?	**C**	The anterior and posterior tibial veins; they run with the arteries of the same name

Lymphatic system · rooms LYMPH4 (set A), LYMPH4B, LYMPH4C

SET A · ROOM LYMPH4 · KEY: CDABADC			ANSWER	
1	CONCEPT	Lymph from the left leg reaches the blood by which route?	C	Thoracic duct into the left subclavian vein
2	ID	Name the structure at the pin.	D	Spleen
3	CONCEPT	Lymph enters a node through many vessels on one side and leaves through few on the other. Which is which?	A	Enters by afferent vessels on the convex side, leaves by efferent vessels at the hilum
4	CONCEPT	Which organ is a primary lymphatic organ, and what makes it primary?	B	Thymus, because T lymphocytes mature there
5	ID	Name the structure at the pin.	A	Pharyngeal tonsil
6	ID	Name the structure at the pin.	D	Lingual tonsil
7	CONCEPT	After lymph nodes are removed from an armpit, the arm on that side swells. What is the disorder and what changed?	C	Lymphedema, lymph cannot drain from the limb

SET B · ROOM LYMPH4B · KEY: CBDADCB			ANSWER	
1	CONCEPT	Lymph moves without a pump. Which three things keep it moving, and in one direction?	C	Skeletal muscle squeezing, breathing pressure changes, and valves in the vessels
2	ID	Name the structure at the pin.	B	Left kidney
3	CONCEPT	What makes a lymphatic capillary different from a blood capillary?	D	It begins as a blind-ended tube with overlapping endothelial flaps that act as one-way minivalves
4	CONCEPT	Which organ is large in a child, shrinks after puberty, and sits in the mediastinum behind the sternum?	A	Thymus
5	ID	Name the structure at the pin.	D	Palatine tonsil
6	ID	Name the structure at the pin.	C	Thymus
7	CONCEPT	A patient's lymph nodes in the neck are swollen and tender during a throat infection. What is the term, and why are they swollen?	B	Lymphadenopathy; lymphocytes multiply in the nodes to fight the infection

SET C · ROOM LYMPH4C · KEY: CABDBAC			ANSWER	
1	CONCEPT	Which quarter of the body drains through the right lymphatic duct, and where does that duct empty?	C	The upper right quarter; the right subclavian vein
2	ID	Name the structure at the pin.	A	Right lobe of liver
3	ID	Name the structure at the pin.	B	Left kidney
4	CONCEPT	In the small intestine the lymphatic capillaries have a special name and a special cargo. What are they, and what do they carry?	D	Lacteals; absorbed fat
5	CONCEPT	The spleen has two kinds of pulp. Which one filters blood and removes old red cells, and which one holds lymphocytes?	B	Red pulp filters blood; white pulp holds lymphocytes
6	ID	Name the structure at the pin.	A	Palatine tonsil
7	CONCEPT	Why is a ruptured spleen an emergency, and what makes it rupture easily?	C	It is full of blood and sits under ribs 9 to 11, so a blow to the left side can tear it and it bleeds heavily